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User-defined element types in `std::simd` through trait-based vectorizable definition

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Abstract

This paper proposes extending `std::simd` to support user-defined element types by replacing the closed list of vectorizable types with a trait-based definition. This minimal change enables type safety, strong typedefs, enumerations, and `std::byte` while maintaining full backward compatibility.

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1. Revision History

1.1. R1 → R2

- Changed approach from customization-focused to trait-based constraints
- Moved customization points to design alternative section
- Provide many implementation examples

1.2. R0 → R1

- Incorporated SG1 and SG6 feedback from 2024 Tokyo meeting
- Added restrictions on element types
- Added inferencing as valid method for constructing simd operators
- Changed from opt-in to opt-out mechanism

2. Introduction

The C++ standard library includes data-parallel types in the `<simd>` header, currently restricting element types to a closed list: arithmetic types and `std::complex` specializations. This paper proposes a minimal change to the specification in which the closed list is replaced with trait-based constraints that handle all existing types while naturally extend support to enumerations, `std::byte` and user-defined types.

Although the change is fairly minimal, this paper thoroughly explores the implications of the changes, including detailed design of type constraints, operator semantics, conversions, and implementation experience. This comprehensive approach is in response to committee feedback requesting evidence that the approach works in practice and careful consideration of edge cases, particularly around type conversions and compiler optimization capabilities.

2.1. Evolution and Design Foundation

Earlier revisions of this proposal focused on providing explicit customization mechanisms for user-defined types. Committee feedback encouraged us to explore element-wise inference instead, making use of the Working Draft's wording in which everything is defined in terms of element operations and element-wise application of those operations. This led to a key question: can modern compilers effectively auto-vectorize element-wise operations on user-defined types? Our investigation showed that leading optimizing compilers can indeed do this remarkably well, and this observation became the foundation of our design.

By relying on compiler optimization, we can open `simd` to user-defined types without requiring customization points for basic operations. This meant we could achieve the desired functionality by simply changing which types are allowed to be elements (i.e., what a vectorizable type is), without modifying operation semantics. The elegance of this approach is that changing only the gate-keeper logic provides the extension we need to support not only user-defined types, but other useful types like enumerations and `std::byte`.

During the last committee meeting, concerns were raised about the performance implications of this approach - what if compilers failed to vectorize the code? To address these concerns we implemented our proposal in Intel's `std::simd` implementation and tested it across multiple generations of Intel architectures with various user-defined types, enumerations, strong typedefs, and specialized DSP types (saturating arithmetic and fixed-point). Implementation experience ([§6 Implementation Experience](#)) demonstrates that with current leading compilers (Clang and Intel oneAPI), these types can generate assembly identical to built-in arithmetic types for standard operations. This proves the approach is viable. Compiler that don't yet optimize as well will improve over time.

2.2. What This Proposal Enables

This proposal allows `simd` to support user-defined types, enumerations, `std::byte`, and other types beyond the current closed list of arithmetic types and `std::complex`. The key requirement is that element-wise application of the scalar operations makes sense for the type.

Examples of types that become vectorizable:

- **User-defined types for type safety:** `struct Meters { float value; };` - strong typedefs that wrap primitives
- **Enumerations:** `enum class Color : uint32_t { Red, Green, Blue };` - type-safe alternatives to raw integers
- **std::byte:** for packet processing and binary data manipulation
- **Specialized arithmetic types:** saturating integers, fixed-point numbers with custom operators
- **Simple aggregates:** `struct RGBA { uint8_t r, g, b, a; };` - small value types with element-wise semantics

What these types share is that the desired SIMD behavior is straightforward: if a type `T` has `operator+`, then `simd<T>` should provide element-wise `operator+` with the same semantics. The scalar operations on `T` define what `simd<T>` should do.

We did find that while element-wise inference works well for most operations (arithmetic, comparisons, permutations, broadcasts), it can occasionally struggle with complex algorithms like reductions or user-defined operators containing branching. To address this, we propose an optional ADL-based customization mechanism (`simd_operator` for operations, `simd_convert` for conversions) that allows users to provide optimized implementations for specific operations while maintaining element-wise inference as the default. This hybrid approach provides a solid foundation that works well in practice while enabling targeted optimization when necessary.

This proposal does NOT address *heterogeneous type operations* where operands have different types and produce a third type (e.g., dimensional analysis where `Length / Time -> Speed`). Such operations represent a fundamentally different design space requiring type-level computation and are explicitly out of scope.

2.3. Core Proposal

The core idea of our proposal is to change definition of a vectorizable type from a closed list to a trait-based definition. A type `T` is now vectorizable if:

- `std::is_trivially_copyable_v<T>` is true
- `sizeof(T)` is 1, 2, 4, 8, or 16
- `alignof(T) <= sizeof(T)`
- `std::disable_vectorization<T>` is false

All existing vectorizable types remain vectorizable with identical semantics. We change only which types are allowed; operator behavior remains element-wise application as currently specified. User-defined types work exactly like arithmetic types of the same size - an operation is available for `vec<T>` if and only if it exists for `T`.

We did notice that it will be necessary to tighten the wording of some operator constraints to explicitly require appropriate return types for user-defined types. This prevents certain classes of errors and performance traps. The constraints distinguish between arithmetic types (which may undergo integer promotion) and user-defined types (which should return the exact type). For example, `uint8_t + uint8_t` produces an `int` due to integer promotion, requiring lenient checking that allows explicit conversion back. In contrast, user-defined type operators must return the correct type directly to prevent subtle bugs. This doesn't affect existing arithmetic types, but ensures user-defined types behave correctly.

Everything else in the proposal stays the same. All operations and their semantics, performance characteristics, ABI selection, and existing code remain unchanged.

2.4. Scope and Future Directions

2.4.1. In Scope: Element-wise Semantics

This proposal maintains exact semantic parity with existing `simd` operations. All operators require operands of `simd<T>` and return `simd<T>` (or `simd_mask<T>` for comparisons), exactly as `simd<int>` does today. The only change is expanding which types `T` are permitted as elements in a `simd`, moving from a closed list of arithmetic types to a trait-based definition.

This design immediately enables important use cases:

- Type-safe dimensional types that maintain scalar semantics
- Enumeration processing
- `std::byte` for binary data processing
- Domain-specific numeric types (saturating, fixed-point, custom precision)
- Future numeric types (bfloat16, float8, and other emerging formats)

Beyond user-defined types, the trait-based approach future-proofs `simd` for numeric type evolution. Compiler builtins or emerging standard types like `std::bfloat16_t` or `std::float8_t`, and vendor-specific formats automatically work without requiring standard amendments. As hardware evolves for machine learning and scientific computing, new numeric types integrate seamlessly into `simd`.

The trait-based gatekeeper change provides substantial value independently, enabling these use cases without requiring the committee to solve significantly harder problems.

2.4.2. Deliberately Out of Scope: Heterogeneous Operations

Heterogeneous type operations, where `simd<A> op simd<B> -> simd<C>`, are explicitly excluded from this proposal. Such operations require fundamentally different design considerations:

1. **Type-level computation:** Result types must be computed from operand types
2. **Operator overload complexity:** Every binary operator needs templates for all valid type combinations
3. **ABI challenges:** If the input types have different ABIs they must somehow be reconciled
4. **Specification burden:** Defining which type combinations are valid and their semantics

More critically, this would be a change to `simd` itself, not just which types participate. Current `simd<int>` only supports homogeneous operations (`simd<int> + simd<int> -> simd<int>`). Extending `simd` to support heterogeneous operations should be proposed separately and would apply to all element types, not just user-defined ones. This proposal does not and should not solve that design problem.

2.4.3. Forward Compatibility

The current design is fully forward-compatible with future heterogeneous operations. Adding template overloads such as:

```
template<typename T, typename U, typename Abi>
friend basic_simd</* computed result type */, Abi>
operator+(const basic_simd<T, Abi>&, const basic_simd<U, Abi>&);
```

would not conflict with existing homogeneous operators - it would simply add new overloads to the existing set. The trait-based vectorizable definition in this proposal works unchanged with such future extensions.

2.4.4. Rationale for Deferral

We defer heterogeneous operations because:

1. **Proven need vs. speculation:** This proposal solves demonstrated problems (type-safe wrappers, enums, byte processing) with implementation experience. No concrete use cases for allowing unit-like operations in `simd` have been presented.
2. **Domain expertise:** Heterogeneous operations should be designed by experts in dimensional analysis and units libraries who understand the requirements. This proposal focuses on transparent wrappers - a simpler, well-understood case with clear use cases and proven implementation.
3. **Implementation burden:** Supporting heterogeneous operations would significantly increase specification and implementation complexity without demonstrated need. Implementations already have freedom to optimize transparent wrapper operations effectively, and our implementation experience shows they do so successfully.

4. **Incremental progress:** Solving the well-understood transparent wrapper case now delivers immediate value. More complex type-algebras can be addressed in future work with actual implementation experience to guide design, should such libraries ever reach maturity.

Implementation experience ([§6 Implementation Experience](#)) demonstrates that user-defined types generate optimal code for common operations, validating this design approach.

§ 3. Motivation

The current restriction to arithmetic types prevents several valuable use cases that would naturally benefit from SIMD parallelism, including strong typedefs for physical units, enumerations for state machines and flags, `std::byte` for low-level data processing, and small compound types for structure-of-arrays patterns. This section presents motivating examples.

§ 3.1. Type Safety and Strong Typedefs

Physical units, identifiers, and other domain-specific types are commonly wrapped in strong typedefs to prevent semantic errors:

```
struct Meters { float value; };
struct Seconds { float value; };

// Type safety at scalar level
Meters distance{100.0f};
Seconds time{5.0f};
// Meters m = time; // Error: type mismatch

// Same type safety should extend to parallel code
vec<Meters> distances = {100.0f, 200.0f, 150.0f, 180.0f};
vec<Seconds> times = {5.0f, 10.0f, 7.5f, 9.0f};
// vec<Meters> m = times; // Should also be error
```

Currently users who wish to put these strong types into a `basic_vec` would need to unpack them to `vec<float>`, losing type safety precisely where parallel operations occur. This proposal preserves type safety uniformly.

§ 3.2. Signal and Media Processing Types

Specialized domains use custom numeric types optimized for their workloads:

```
// Fixed-point arithmetic for digital signal processing
struct fixed_point_16s8 {
    std::int16_t data;

    fixed_point_16s8 operator+(fixed_point_16s8 rhs) const {
        return fixed_point_16s8{saturate_add(data, rhs.data)};
    }
    // Other operators...
};

// Should work with vec
vec<fixed_point_16s8> samples = load_audio_samples();
auto processed = apply_filter(samples); // Element-wise fixed-point operations
```

The proposal allows `std::simd` to provide its parallel infrastructure (loads, stores, masking, permutations, reductions) while deferring arithmetic to the user-defined type's operators.

3.3. Enumerations

Enumerations are essentially only restricted integer types with named values. They are widely used for state machines, flags, and encoded data. Vectorizing enumerations enables batch processing of such data.

```
enum class Color : std::uint32_t { Red, Green, Blue, Alpha };

vec<Color> pixel_channels = /* ... */;
auto masked = pixel_channels & Color::Alpha; // Bitwise operations on scoped enums
```

Scoped enums (`enum class`) only allow operations that are valid for the enum itself (typically bitwise operations, comparisons, and conversions), while unscoped enums allow arithmetic operations through implicit conversion to their underlying type. The element-wise application mechanism automatically respects these restrictions without any special handling.

Batch processing of enumeration values is useful for state machines, flags, and encoded data.

3.4. `std::byte`

`std::byte` is a distinct type representing raw byte data, commonly used in low-level programming. Vectorizing `std::byte` enables efficient byte-level operations such as encryption, checksums, and encoding.

```
vec<std::byte> data = /* load from buffer */;
auto encrypted = data ^ vec<std::byte>{0xFF}; // XOR cipher
```

3.5. Compound Types

Small compound types that fit in 16 bytes can be vectorized as atomic units, enabling structure-of-arrays patterns, or packet processing of multiple values simultaneously.

```
// Coordinate pairs
vec<std::pair<int, int>> coordinates;

// RGBA color pixels
vec<std::array<std::uint8_t, 4>> pixels;
```

4. Understanding Type Constraints

To ensure user-defined types work correctly with `std::simd`, we impose constraints that match hardware capabilities and prevent subtle bugs. In summary the constraints are:

- Trivially copyable
- Size: must be 1, 2, 4, 8, or 16 bytes
- Alignment: `alignof(T) <= sizeof(T)`
- Opt-out mechanism via `disable_vectorization`
- Banned standard library types and categories (pointers, unions, cv-qualified, empty)

We now look in more detail at each of these constraints.

4.1. Trivially Copyable Constraint

We require `std::is_trivially_copyable_v<T>`. Many `std::simd` operations move elements bitwise (permutations, broadcasts, gathers, scatters). For these to work correctly, an element's value must be preserved when its bit pattern is copied. Trivially copyable types have no special copy, move, or destroy logic, so bitwise copying always produces correct results.

4.2. Size Constraint

We require `sizeof(T)` to be exactly 1, 2, 4, 8, or 16 bytes. All known hardware vector instruction sets support only power-of-2 element sizes. The largest current vectorizable type is `std::complex<double>` at 16 bytes.

4.3. Alignment Constraint

We require `alignof(T) <= sizeof(T)`. Types with `alignof > sizeof` are excluded as a conservative measure. Over-alignment typically indicates special requirements or semantics beyond simple bitwise operations. For example, hardware-specific alignment requirements, cache-line alignment for lock-free atomics, or other unusual properties. Such types are outside the scope of simple element-wise SIMD semantics and are excluded to avoid complexity and potential misuse.

Exclusion with this constrain is likely to be rare in practice but prevents edge cases with types that have special semantic requirements.

4.4. Padding and Bit Representation

SIMD operations treat element types as uninterpreted bit patterns of the specified size. If a user-defined type contains padding bytes (e.g., `struct ThreeChars { char a, b, c; }` typically has `sizeof=4` with one padding byte), `simd` is agnostic to which bits represent data versus padding. All bits are preserved through operations, with semantics determined solely by the element type's operators. This is consistent with trivially copyable semantics.

4.5. Opt-Out Mechanism

The standard library uses a common pattern for selectively disabling features where a variable template can be specialized. For `std::simd`, this proposal adds `std::disable_vectorization<T>`, which will default to `false` but can be specialized to `true` for types that should not be vectorizable. This mechanism will allow the implementation to opt out of allowing vectorisation for semantically inappropriate types which otherwise appear to permit vectorisation.

Users may specialize `disable_vectorization` for their own types, such as:

```
namespace my_lib {
    struct InternalType { std::uint64_t data; };
}

template<>
inline constexpr bool std::disable_vectorization<my_lib::InternalType> = true;
```

Specializations for cv-qualified or reference types are ill-formed.

4.6. Banned Standard Library Types

In addition to allowing the user to opt out of some types, the mechanism can also be used by the implementation to ban specific standard types and categories which have no meaningful vectorization semantics.

Type categories automatically banned:

- **Pointer types** (`is_pointer_v<T>` or `is_member_pointer_v<T>`): Pointer arithmetic has unclear semantics in SIMD context.
- **Union types** (`is_union_v<T>`): Ambiguity about which member is active.
- **CV-qualified types** (`is_const_v<T>` or `is_volatile_v<T>`): Breaks assignment operators. (Note: cv-qualified vec objects like `const vec<int>` are permitted; the ban applies only to cv-qualified element types.)
- **Empty types** (`is_empty_v<T>`): Carry no data.

Standard library types:

```
// Tag types and sentinels
template<> inline constexpr bool disable_vectorization<std::monostate> = true;
template<> inline constexpr bool disable_vectorization<std::nullptr_t> = true;
template<> inline constexpr bool disable_vectorization<std::nullopt_t> = true;
template<> inline constexpr bool disable_vectorization<std::in_place_t> = true;
template<> inline constexpr bool disable_vectorization<std::allocator_arg_t> = true;
template<> inline constexpr bool disable_vectorization<std::piecewise_construct_t> = true;

// Compile-time types
template<> inline constexpr bool disable_vectorization<std::source_location> = true;
template<class T, T v> inline constexpr bool
    disable_vectorization<std::integral_constant<T, v>> = true;

// Nested simd types
template<class T, class Abi>
inline constexpr bool disable_vectorization<std::basic_vec<T, Abi>> = true;
template<class T, class Abi>
inline constexpr bool disable_vectorization<std::basic_mask<T, Abi>> = true;
```

Note that under these constraints, arrays (`int[4]`), `std::pair`, and `std::tuple` are not banned, provided they satisfy the constraints. They can all be useful in their own way, such as representing vector-processing of packet processing patterns, structured data, and structure-of-array layouts. Even if these types do not provide arithmetic or mathematical operations, it is still useful to be able to use them for parallel load/store, masking, permutation and bit-level operations.

This list is not exhaustive; implementations may provide additional specializations for other types where vectorization is semantically inappropriate.

4.7. Summary of Constraints

The constraints work together to ensure types are safe and efficient for vectorization:

- **Trivially copyable** enables bitwise element manipulation
- **Power-of-2 size** matches hardware vector capabilities
- **Alignment constraint** prevents types with special semantic requirements
- **Opt-out mechanism** allows excluding inappropriate types

These enable user-defined types like `vec<Meters>`, `vec<Color>`, `vec<std::byte>`, and `vec<std::array<uint8_t, 4>>`, while excluding pointers, unions, cv-qualified types, empty types, and opted-out types.

5. Operations on User-Defined Types

This section describes how `std::simd` operations work with user-defined element types. The key principle is element-wise application: operations on `vec<T>` apply the corresponding operation on `T` to each element independently.

User-defined types are treated as atomic blocks of bits whose internal structure is not modified by `simd` operations. This proposal does not include struct-of-arrays conversions or layout transformations for user-defined types.

5.1. Operator Constraints

The `std::simd` specification provides operators conditionally using `requires` clauses. The working draft currently checks only that element-wise operations are valid expressions, without constraining return types. This proposal tightens these constraints to require appropriate return types, with different rules for arithmetic types versus user-defined types.

For **arithmetic types and unscoped enumerations**, operators may return a promoted type (e.g., `uint8_t + uint8_t → int`), which is then explicitly converted back to the element type. This preserves existing behavior for built-in types.

For **all other types** (scoped enumerations, `std::byte`, `std::complex`, and user-defined types), operators must return exactly `value_type`. This prevents subtle bugs where user-defined operators return incorrect types.

The constraints use exposition-only concepts that capture this two-tier checking:

```
template<typename T, typename BinaryOp>
concept supported-binary-op = /* exposition only */
    (is_arithmetic_v<T> || (is_enum_v<T> && !is_scoped_enum_v<T>)) ?
    requires(T a, T b) { T(BinaryOp{}(a, b)); } :
    requires(T a, T b) { { BinaryOp{}(a, b) } -> same_as<T>; };
```

Return type requirements:

Arithmetic operators (+, -, *, /, %, &, |, ^, <<, >>, unary -, ~):

- For arithmetic types and unscoped enums: Allow promotion with explicit conversion back
- For all other types: Must return exactly `value_type`

Comparison operators (==, !=, <, <=, >, >=):

- Must return `bool` (no promotion of result type)

These requirements prevent size mismatches, avoid conversions that change semantics, and prevent performance traps from proxy types, while maintaining backward compatibility for arithmetic types.

NOTE: Comparison operators are not synthesized from each other, maintaining parity with existing `simd` behavior for arithmetic types. For example, `operator!=` is not synthesized from `operator==`. This avoids introducing inconsistency with current `simd` semantics. Synthesis of comparison operators could be proposed separately as an enhancement to all `simd` types (including arithmetic types), not just user-defined ones.

Examples:

```
struct Meters {
    float value;
    Meters operator+(Meters rhs) const { return Meters{value + rhs.value}; }
    bool operator<(Meters rhs) const { return value < rhs.value; }
};

vec<Meters> a, b;
auto sum = a + b;           // ✓ OK: operator+ returns Meters
auto mask = a < b;          // ✓ OK: operator< returns bool

struct NoAdd { float value; };
vec<NoAdd> x, y;
auto result = x + y;        // ✗ Error: operator+ not defined

struct DifferentReturn {
    int16_t value;
    int32_t operator+(DifferentReturn) const; // Change return type
};
vec<DifferentReturn> v, w;
auto bad = v + w;          // ✗ Error: int32_t is not DifferentReturn
```

Compound assignments use the same constraints as their corresponding binary operators:

```
friend constexpr basic_simd& operator+=(basic_simd& lhs, const basic_simd& rhs)
    requires supported-binary-op<value_type, plus<>>; // Same as operator+
```

All six comparison operators continue to be independently specified.

The mask type `basic_mask<value_type, Abi>` is determined by the element type's size, not its contents. Masks indicate active/inactive lanes for a group of bits of size `sizeof(value_type)`. For any user-defined type, the mask semantics

are identical to those of arithmetic types of the same size - one mask bit per element, regardless of what data the element contains.

5.2. Conversions and Casts

Converting constructors use `static_cast` for element conversion:

```
//Element `i` is initialized with `static_cast<T>(v[i])`.
template<typename U>
explicit constexpr basic_vec(const basic_vec<U, Abi>& v)
    requires /* appropriate constraints */;
```

This naturally supports user-defined conversions:

```
struct Meters { float value; };
struct Feet {
    float value;
    operator Meters() const { return Meters{value * 0.3048}; }
};

vec<Feet> feet = {3.0f, 6.0f, 9.0f, 12.0f};
vec<Meters> meters{feet}; //  Works via conversion operator
```

The existing `static_cast` semantics handle all conversion scenarios without additional specification.

5.2.1. Value-Preserving Conversions

The working draft defines "value-preserving" only for conversions from arithmetic types: *"The conversion from an arithmetic type U to a vectorizable type T is value-preserving if all possible values of U can be represented with type T "* ([[simd.general](https://eel.is/c++draft/simd#general-8)](https://eel.is/c++draft/simd#general-8)). This definition is precise for arithmetic types but does not extend to user-defined types.

For conversions involving user-defined types, this proposal defers to the type author's judgment as expressed through implicit versus explicit conversions:

- For arithmetic-to-arithmetic conversions: Use the existing value-preserving definition (e.g., `int` to `long` is value-preserving, but `double` to `float` is not).

For conversions involving at least one user-defined type: Use `std::is_convertible_v<From, To>` to determine if the conversion may be implicit:

- If `is_convertible_v<From, To>` is `true`, the type author has declared the conversion safe via an implicit constructor, so `simd` allows it implicitly
- If `is_convertible_v<From, To>` is `false` but `is_constructible_v<To, From>` is `true`, the type author requires `explicit`, so `simd` also requires explicit conversion

Examples:

```
struct Meters {
    float value;
    Meters(float f) : value(f) {} // Implicit - author says it's safe
};

struct Feet {
    float value;
    explicit Feet(float f) : value(f) {} // Explicit - author says be careful
};

vec<float> vf = {...};
```

```

vec<Meters> v0 = vf;           // OK - Meters(float) is implicit
vec<Feet> v1 = vf;            // Error - Feet(float) is explicit
vec<Feet> v2 = vec<Feet>(vf); // OK - explicit construction

std::span<float, 1024> sf;

// OK - implicit conversion from float to Meters
auto m_vec = unchecked_load<vec<Meters, 8>>(sf);

// Error - implicit conversion from float to Feet not allowed
auto f_vec = unchecked_load<vec<Feet, 8>>(sf);

// OK - conversion from float allowed with flag_convert tag
auto f_vec = unchecked_load<vec<Feet, 8>>(sf, flag_convert);

```

This approach:

- Respects the type author's design decisions about safety
- Maintains consistency with scalar usage patterns
- Avoids second-guessing the type author's judgment
- Does not require `simd` to define value-preservation semantics for user-defined types

Same-type operations are unaffected - `simd<Meters>(Meters{3.14f})` broadcasts by copying, not converting, so these rules don't apply.

Note that a type author could declare an implicit conversion that loses information (e.g., `Meters(double)` with `float` storage). However, this is the type author's choice at the scalar level, and `simd` should not override that judgment. If the scalar user-define type allows implicit lossy conversion, `simd` does too.

5.3. Reductions

Reduction operations (e.g. `reduce`, `reduce_min`, `reduce_max`) apply the operation pairwise:

```

// Applies `binary_op` pairwise to elements in unspecified order.
template<typename T, typename Abi, typename BinaryOp = std::plus<>>
constexpr T reduce(const basic_vec<T, Abi>& v, BinaryOp binary_op = {});

```

NOTE: Reductions assume associativity. For types with non-associative operations, results may differ from sequential left-to-right reduction. This is consistent with floating-point behavior, where `reduce(v, std::plus<>())` may produce different results than sequential summation due to intermediate rounding. The working draft already specifies this behavior via preconditions on the binary operation.

```

struct ModularInt {
    int value;
    ModularInt operator+(ModularInt rhs) const {
        return ModularInt{(value + rhs.value) % 100};
    }
};

vec<ModularInt> v = {50, 30, 40, 20};
auto sum = reduce(v, std::plus<>());
// Result: ModularInt{40}
// Could evaluate as
// ((50+30)+40)+20 = (80+40)+20 = 20+20 = 40
// (50+30)+(40+20) = 80+60 = 40

```

5.4. Maths Functions

Maths functions like `sin`, `cos`, `sqrt`, `exp`, etc. are constrained to arithmetic types in the working draft. For user-defined types, these functions are not automatically provided and attempting to use them results in a compile error. This is to avoid any accidental performance cliffs from naive element-wise implementations. Unlike arithmetic operators, which are typically simple in most cases, maths functions are more likely to have complex implementations that are not suitable for element-wise application:

```
vec<MyFloat> v;  
auto result = sin(v); // ❌ Compile error: constrained to arithmetic types
```

Users must provide explicit overloads via ADL if they want these functions for their types:

```
template<typename Abi>  
basic_vec<MyFloat, Abi> sin(const basic_vec<MyFloat, Abi>& v) {  
    // User-provided vectorized implementation  
}
```

NOTE: The functions `min` and `max` are different from other mathematical functions. The working draft defines them as element-wise operations that call the element type's `min/max` function (found via ADL) or use `operator<` for comparison. These work automatically for user-defined types that provide the necessary operations, as demonstrated in the implementation experience section.

5.5. Load and Store Operations

Load operations already specify element conversion via `static_cast`:

```
// Element `i` is initialized with `static_cast<T>(*std::next(first, i))`.  
template<typename It>  
constexpr basic_vec(It first, It last);
```

This naturally handles both same-type loads and converting loads via the `static_cast` mechanism (see [§5.2 Conversions and Casts](#) for examples). Implementations may optimize by using vector loads followed by vector conversions rather than converting each element individually.

Store operations work similarly. No specification changes are needed.

5.6. Copy Operations

Operations that move elements without interpreting values work on any trivially copyable type:

- `permute`, `broadcast` - rearrange elements
- `compress`, `expand` - conditional packing/unpacking
- `select` - conditional element selection
- `chunk`, `cat` - size/shape changes

These operate at the bit level and require no knowledge of element semantics. The trivially copyable constraint ensures they already work correctly for user-defined types.

5.7. Implementation Considerations

In this section we shall briefly examine two important implementation considerations when supporting user-defined types in `std::simd`: exception safety and ABI selection.

All `basic_simd` operations are declared `noexcept` in the working draft. This has important implications for user-defined types: if an element-wise operation throws an exception during a `simd` operation, `std::terminate` will be called.

This behavior is appropriate for SIMD code. Detecting and propagating exceptions on individual elements would require serializing the operation, checking each element's result, and managing partial completion state. This fundamentally contradicts SIMD's purpose of parallel execution. User-defined types intended for use in `simd` should have non-throwing operations, or accept that exceptions will terminate the program.

The `noexcept` specification means:

- Element-wise operations are not required to be `noexcept` themselves
- If they do throw during `simd` operations, `std::terminate` is called
- Users must ensure their types' operations don't throw in practice
- This is consistent with SIMD being performance-critical code where exceptions are inappropriate

§ 5.8. ABI Selection for User-Defined Types

ABI selection determines the vector width (number of elements) for a `simd` object. For user-defined types, ABI selection is based solely on `sizeof(T)`. A UDT of size N bytes is treated identically to arithmetic types of size N for ABI purposes.

The `alignof(T) <= sizeof(T)` constraint ensures compatible memory layout, but `alignof(T)` does not influence ABI selection. This means:

```
struct A { int32_t x; }; // sizeof=4 → treated like int32_t for ABI
struct B { float f; }; // sizeof=4 → treated like float for ABI
struct C { uint8_t data[4] }; // sizeof=4, alignof=1 → treated like int32_t for ABI
```

Any two types with the same size will receive the same ABI and therefore the same number of elements:

```
struct MyInt32 { std::int32_t value; };

vec<int> v1; // Suppose this gets 512-bit vectors = 16 elements
vec<float> v2; // Also 512-bit vectors = 16 elements (both 4 bytes)
vec<MyInt32> v3; // Also 512-bit vectors = 16 elements (also 4 bytes)
```

Implementations select vector width based on element size to match hardware capabilities. This ensures consistent behavior and predictable performance characteristics across types of the same size.

§ 6. Implementation Experience

We implemented this approach in Intel's `std::simd` implementation and tested across multiple Intel architectures. This section presents the technical details: code generation results, assembly analysis, and identified limitations.

§ 6.1. Test Implementation

We experimented with a number of different test types, including an enumeration, a strong type, and a saturating integer type to evaluate code generation quality:

```
enum Color {Red, Green, Blue};

struct Meters {
    float value;

    Meters operator+(Meters rhs) const { return Meters{value + rhs.value}; }
    bool operator<(Meters rhs) const { return value < rhs.value; }
};
```

```

struct saturating_int16 {
    saturating_int16(int v) : data(v) {}
    std::int16_t data;

    // Saturating addition
    friend saturating_int16 operator+(saturating_int16 lhs, saturating_int16 rhs) {
        auto r = std::int32_t(lhs.data) + std::int32_t(rhs.data);
        return saturating_int16(std::clamp<int32_t>(r, -32768, 32767));
    }

    friend bool operator>(saturating_int16 lhs, saturating_int16 rhs) {
        return lhs.data > rhs.data;
    }

    // Other operators defined similarly...
};

```

6.2. Successful Inference Cases

Testing was performed with Clang 20 and Intel oneAPI 2025.0 targeting Intel Sapphire Rapids. For most operations, these compilers generated excellent code from element-wise operator application. The generated assembly uses native vector instructions throughout, with no scalar fallback or element-by-element processing. The instruction selection matches what hand-written intrinsics would produce, demonstrating that element-wise inference can generate performance-competitive code for common operations.

Important note on compiler variance: Optimization quality for user-defined types varies significantly between compiler vendors and versions. The results presented here reflect what’s possible with current leading implementations - other compilers may produce substantially less optimal code, particularly for complex operations like reductions. This variance is a quality-of-implementation issue, not a fundamental limitation of the design. Clang and oneAPI demonstrate the approach works. Compilers that currently struggle will improve over time as their optimization passes mature. Users should verify code quality with their specific toolchains and consider using the optional customization mechanisms ([§7 Design Alternative: Customization Points](#)) if their compiler doesn’t yet optimize well.

See [§13 Appendix: Assembly Code Examples](#) for detailed assembly listings showing the code generated for a variety of common patterns.

6.3. Identified Limitation

We did identify one case where element-wise inference produced suboptimal code:

C++ Code	Generated Assembly (Suboptimal)
<pre> auto reduce_add(vec<saturating_int16> v) { return reduce(v, std::plus<>{}); } </pre>	<pre> reduce_add(...): vextracti128 xmm1, ymm0, 1 vpaddsw xmm0, xmm0, xmm1 vpextrq rdx, xmm0, 1 vmovq rax, xmm0 mov rsi, rax shr rsi, 48 mov rcx, rdx shr rcx, 48 lea edi, [rsi + rcx] movsx edi, di sar edi, 15 xor edi, -32768 add si, cx cmovo esi, edi // ... continues with scalar operations </pre>

For this reduction, the compiler started with vector operations but then switched to element-by-element scalar execution. The first two instructions are correct (extract and vector add), but subsequent operations process elements individually rather than maintaining vectorization throughout.

6.4. Implications for Customization

This experience demonstrates that:

1. Element-wise inference succeeds for most operations with leading compilers: Permutations, broadcasts, and direct operators generate optimal code with current Clang and Intel oneAPI implementations.
2. Compiler maturity varies significantly: Optimization quality for user-defined types shows substantial differences between compiler vendors and versions. While Clang and oneAPI generate excellent code, other compilers may produce significantly less optimal results - sometimes falling back to scalar operations where vectorization should succeed. This reflects differences in compiler optimization sophistication, not limitations of the design itself.
3. Specific limitations exist: Even with mature compilers, complex algorithms like reductions may not auto-vectorize perfectly from scalar operator definitions.
4. Customization provides value: For cases where compilers struggle, the ADL-based customization mechanism (`simd_operator` and `simd_convert`) enables users to provide optimized implementations, ensuring good performance regardless of compiler optimization quality.

The identified limitations motivated the customization design presented in [§7 Design Alternative: Customization Points](#). However, these limitations do not diminish the value of the core proposal's element-wise inference and the customization mechanism serves as both a performance optimization for complex cases and a portability tool for users working with compilers that haven't yet achieved sophisticated UDT vectorization.

6.5. Implementation Impact

Implementations already handle element types generically for many operations (permutations, broadcasts, masking). The trait-based definition formalizes this practice and extends it uniformly.

The following changes are needed:

- Modify type trait checking for the `vectorizable` concept/trait
- Add `disable_vectorization` variable template with standard library specializations
- Update operator constraints to check return types (constraints already present, only need tightening)

The effort to customize the implementation is minimal. The core machinery already exists and only the gate-keeping logic changes. The implementation experience demonstrates the approach described in this proposal is viable.

7. Design Alternative: Customization Points

Implementation experience demonstrated that element-wise inference produces correct, performant code for most operations. However, we identified cases where reductions did not generate optimal code (see [§6.3 Identified Limitation](#)), and users may have types with complex operators that inhibit compiler vectorization. For such cases, optional customization points allow users to provide optimized implementations.

We propose two ADL-discovered customization points:

Operations customization: A single overloaded function `simd_operator` handles unary and binary operations:

```
// In user's namespace:
auto simd_operator(vec<T> v, Op op) -> vec<T>;           // Unary
auto simd_operator(vec<T> v1, vec<T> v2, Op op) -> vec<T>; // Binary

// Although not needed yet, ternary operations would naturally be handled too.
auto simd_operator(vec<T> a, vec<T> b, vec<T> c, Op op) -> vec<T>;
```


Conversion customization: A separate function handles type conversions using a tag-based dispatch pattern. The `convert_to_t<T>` class template and `convert_to<T>` variable template are provided as part of the public API:

```
// Provided by the simd library:
template<typename T>
struct convert_to_t {
    using type = T;
    constexpr explicit convert_to_t() noexcept = default;
};

template<class T> inline constexpr convert_to_t<T> convert_to{};
```

The user can then provide overloads of `simd_convert` for specific type conversions:

```
// User customization point signature:
template<typename Abi>
basic_vec<To, Abi> simd_convert(const basic_vec<From, Abi>& source, convert_to_t<To>);
```

The `convert_to_t<T>` tag argument serves two purposes: it enables ADL discovery (since the destination type would otherwise only appear as a template parameter), and it allows users to write customization points for specific conversion directions.

Conversion dispatch: When a `simd` conversion is needed, the implementation uses a three-tier dispatch strategy:

1. **Arithmetic types:** If both the source and destination element types are arithmetic types (including `std::complex`), the implementation uses its own optimized conversion (e.g., compiler builtins). The `simd_convert` customization point is never checked. This prevents the user from accidentally overriding well-optimized conversions for arithmetic types, which are common and performance-critical.
2. ****ADL `simd_convert`**:** If at least one type is not arithmetic, the implementation checks whether a user-provided `simd_convert` customization point exists via ADL. If found, and it returns exactly `basic_vec<To, Abi>`, it is used.
3. **Element-wise fallback:** If no `simd_convert` customization is found, the implementation falls back to element-wise `static_cast`, which invokes the scalar conversion operators or constructors on each element.

Separate dispatch paths for operations: The standard distinguishes between types that must always use optimized implementations and types that may provide customization. The following types never check for customization and always use optimized implementations from the library:

- Types where `std::is_arithmetic_v<T>` is `true` (all arithmetic types)
- `std::byte`
- Specializations of `std::complex`

All other vectorizable types (including enumerations and user-defined types) check for customization points via ADL:

```
// Conceptual specification
template<typename T> // Arithmetic types, std::byte, std::complex
    requires std::is_arithmetic_v<T> || std::is_same_v<T, std::byte> || /* complex */
friend basic_vec operator+(const basic_vec& lhs, const basic_vec& rhs) {
    return /* implementation-defined optimized implementation */;
}

template<typename T> // Enumerations and user-defined types
    requires (!std::is_arithmetic_v<T> && !std::is_same_v<T, std::byte> && /* not complex */)
friend basic_vec operator+(const basic_vec& lhs, const basic_vec& rhs) {
    if constexpr (requires { simd_operator<T>(lhs, rhs, std::plus{}); }) {
        return simd_operator<T>(lhs, rhs, std::plus{}); // Custom via ADL
    } else {
        return /* element-wise application */; // Default
    }
}
```

For enumerations and user-defined types without customization, the `simd_operator` check fails at compile time and element-wise inference is used. Since enumerations without custom operators compile to simple integer arithmetic, element-wise inference produces optimal code.

Example: A user can provide `simd_operator(vec<saturating_int16>, vec<saturating_int16>, std::plus<>)` to use native saturating add instructions, while relying on element-wise inference for all other operations.

Committee guidance requested: Should customization points be included in P2964 or deferred to a separate paper? The core proposal provides correct semantics and reasonable performance without them. Customization enhances optimization but is not essential for functionality.

Technical details are provided in [§12 Appendix: Customization Point Technical Details](#).

§ 8. Design Options for Enum and Byte Support

With our proposal, enumerations and `std::byte` now become vectorizable. Consequently, related utility functions could be extended to work with `simd`:

```
// Element-wise to_underlying for enumerations
template<class Enum, class Abi>
constexpr rebind_t<underlying_type_t<Enum>, basic_vec<Enum, Abi>>
    to_underlying(const basic_vec<Enum, Abi>& v) noexcept;

// Element-wise to_integer for std::byte
template<class IntegerType, class Abi>
constexpr rebind_t<IntegerType, basic_vec<byte, Abi>>
    to_integer(const basic_vec<byte, Abi>& v) noexcept;
```

These provide consistency with their scalar counterparts and convenience for common conversions. However, they are not strictly necessary for this paper, and could be introduced as a paper in their own right at a later stage.

Committee guidance requested: Should these utilities be (1) included in this proposal for completeness, (2) deferred to a separate proposal focused on convenience utilities, or (3) omitted entirely? Optional wording is provided in [§9.11 \(OPTIONAL\) Add overload for to_underlying](#) and [§9.12 \(OPTIONAL\) Add overload for to_integer](#).

§ 9. Proposed Wording

The wording in this section is relative to the working draft at <https://eel.is/c++draft/simd>.

§ 9.1. Modify [simd.general]

Modify [simd.general] as follows:

~~The set of *vectorizable types* comprises~~

- ~~all standard integer types, character types, and the types `float` and `double`;~~
- ~~`std::float16_t`, `std::float32_t`, and `std::float64_t` if defined; and~~
- ~~`complex<T>` where `T` is a vectorizable floating-point type.~~

A type `T` is a *vectorizable type* if:

- `is_trivially_copyable_v<T>` is true
- `sizeof(T)` is 1, 2, 4, 8, or 16
- `alignof(T)` is less than or equal to `sizeof(T)`
- `disable_vectorization<T>` (see [simd.disable]) evaluates to false.

9.2. Add `[simd.disable]` after `[simd.general]`

Insert a new subclause **[simd.disable]** after `[simd.general]`:

9.2.1. Disabling vectorization `[simd.disable]`

```
template<class T>
inline constexpr bool disable_vectorization = see below;
```

The variable template `disable_vectorization<T>` evaluates to `true` if any of the following conditions hold:

- `is_pointer_v<T>` is `true`, or
- `is_member_pointer_v<T>` is `true`, or
- `is_union_v<T>` is `true`, or
- `is_const_v<T>` is `true`, or
- `is_volatile_v<T>` is `true`, or
- `is_empty_v<T>` is `true`, or
- A program-defined or implementation-provided specialization of `disable_vectorization<T>` explicitly sets it to `true`.

Otherwise, `disable_vectorization<T>` evaluates to `false`.

A program may provide explicit specializations of `disable_vectorization` for program-defined types. Such specializations shall be usable in constant expressions and have type `constexpr bool`.

Specializations of `disable_vectorization` for cv-qualified types or reference types are ill-formed.

The implementation provides explicit specializations that set `disable_vectorization` to `true` for the following standard library types: `monostate`, `nullptr_t`, `nullopt_t`, `in_place_t`, `allocator_arg_t`, `piecewise_construct_t`, `source_location`, `integral_constant<T, v>`, `basic_simd<T, Abi>`, and `basic_simd_mask<T, Abi>`.

Implementations may provide additional specializations for other types where vectorization is semantically inappropriate.

9.3. Add exposition-only concepts to [simd.expos]

Add the following to [simd.expos], after the existing exposition-only definitions:

```
template<typename T>
concept promotable_type = // exposition only
    is_arithmetic_v<T> || (is_enum_v<T> && !is_scoped_enum_v<T>);

template<typename T, typename UnaryOp>
concept supported_unary_op = // exposition only
    promotable_type<T> ?
    requires(T a) { UnaryOp{}(a); } :
    requires(T a) { { UnaryOp{}(a) } -> same_as<T>; };

template<class T, class BinaryOp>
concept supported_binary_op = // exposition only
    ( promotable_type<T> && requires(T a, T b) { BinaryOp{}(a, b); }) ||
    (!promotable_type<T> && requires(T a, T b) { BinaryOp{}(a, b) -> std::same_as<T>; });
```

[Note: The `promotable_type` concept identifies types that participate in C++'s standard implicit conversion and integer promotion rules (arithmetic types and unscoped enumerations). For these types, binary operations may return a promoted type that requires explicit conversion back to `value_type` (e.g., `uint8_t + uint8_t` returns `int`). For all other vectorizable types (scoped enumerations, `std::byte`, `std::complex`, and user-defined types), operations must return exactly `value_type`. —end note]

9.4. Modify [simd.ctor] broadcasting constructor

Modify the constraints for the broadcasting constructor `explicit constexpr basic_simd(value_type x)` in [simd.ctor]:

Constraints:

- ~~From is not an arithmetic type and does not satisfy constexpr-wrapper-like, or~~
- `From` is not an arithmetic type and does not satisfy `constexpr-wrapper-like` and `is_convertible_v<From, value_type>` is true, or
- `From` is an arithmetic type and the conversion from `From` to `value_type` is value-preserving ([simd.general]), or
- `From` satisfies `constexpr-wrapper-like`, `remove_cvref_t<decltype(From::value)>` is an arithmetic type, and `From::value` is representable by `value_type`.

Drafting note: This ensures that conversions involving user-defined types respect the type author's design. If the scalar type requires explicit conversion (e.g., `explicit Meters(float)`), the `simd` conversion also requires explicit construction. If the scalar type allows implicit conversion, `simd` follows suit.

9.5. Modify [simd.ctor] converting constructor

Modify the Remarks paragraph for the converting constructor `template<class U, class UAbi> explicit(see below) basic_simd(const basic_simd<U, UAbi>& x)` in [simd.ctor]:

Remarks: The expression inside `explicit` evaluates to `true` if any of the following hold:

Modify the first condition (about value-preserving) to clarify it only applies when both types are arithmetic:

- ~~the conversion from `U` to `value_type` is not value-preserving, or~~
- both `U` and `value_type` are arithmetic types and the conversion from `U` to `value_type` is not value-preserving, or

Add a new condition after the value-preserving check:

- at least one of `U` or `value_type` is not an arithmetic type and `is_convertible_v<U, value_type>` is false, or

The remaining conditions about integer conversion rank and floating-point conversion rank remain unchanged.

Drafting note: This extends the explicit-ness determination to user-defined types. For UDT conversions, we check `is_convertible_v` rather than value-preserving (which is only defined for arithmetic types).

The phrase "at least one of `U` or `value_type` is not an arithmetic type" covers three cases:

- Arithmetic** $\rightarrow$ **UDT** (e.g., `simd<float>` $\rightarrow$ `simd<Meters>`): requires and respects whether the UDT provides an implicit or explicit constructor from the arithmetic type (e.g., `Meters(float)`)
- UDT** $\rightarrow$ **Arithmetic** (e.g., `simd<Meters>` $\rightarrow$ `simd<float>`): requires and respects whether the UDT provides an implicit or explicit conversion operator to the arithmetic type (e.g., `operator float()`)
- UDT** $\rightarrow$ **UDT** (e.g., `simd<Meters>` $\rightarrow$ `simd<Feet>`): requires and respects whether conversion is available and whether it's implicit or explicit. The UDT author can provide either a constructor in the target type (`Feet(Meters)`) or a conversion operator in the source type (`Meters::operator Feet()`) allowing `static_cast` to use whichever is available.

The arithmetic $\rightarrow$ arithmetic case is handled by the first condition's value-preserving check. We use "at least one" (not "both") because we want to respect the type author's implicit/explicit judgment for *any* conversion involving a UDT. The `is_convertible_v` check will fail (requiring explicit construction) if the necessary constructor or conversion operator doesn't exist or is marked explicit.

§ 9.6. Modify [simd.binary]

Modify the constraints in [simd.binary] as follows:

Let *op* be the operator.

Constraints: ~~requires (value_type a, value_type b) { a op b; } is true~~ supported-binary-op<value_type, Op> is true, where Op is the corresponding standard transparent function object (plus<>, minus<>, multiplies<>, divides<>, modulus<>, bit_and<>, bit_or<>, bit_xor<>).

Returns: A `basic_simd` object initialized with the results of applying *op* to *lhs* and *rhs* as a binary element-wise operation.

For the shift operators:

Let *op* be the operator.

Constraints: ~~requires (value_type a, simd_size_type b) { a op b; } is true~~ supported-binary-op<value_type, Op> is true, where Op is the corresponding standard transparent function object.

Note: [P4006] proposes adding `bit_lshift<>` and `bit_rshift<>` function objects for the shift operators. The C++ standard currently lacks transparent function objects for shift operators, which would provide cleaner specification of shift behavior (see [P4006]). However, this proposal specifies shift operator behavior directly using constraint-based semantics and is independent of P4006. If P4006 is adopted, it would provide an alternative specification approach but does not affect the functionality proposed here.

9.7. Modify [simd.cassign]

Modify the constraints in [simd.cassign] as follows:

Let *op* be the operator.

Constraints: ~~requires (value_type a, value_type b) { a op b; } is true~~ supported-binary-op<value_type, Op> is true, where Op is the standard transparent function object corresponding to the binary operator with the same name (e.g., operator+= uses the constraint from operator+).

For the shift compound assignment operators:

Let *op* be the operator.

Constraints: ~~requires (value_type a, simd_size_type b) { a op b; } is true~~ supported-binary-op<value_type, Op> is true, where Op is the standard transparent function object corresponding to the binary operator.

9.8. Modify [simd.comparison]

Modify the constraints in [simd.comparison] as follows:

Let *op* be the operator.

Constraints: ~~requires (value_type a, value_type b) { a op b; } is true~~ requires (value_type a, value_type b) { { a op b } -> same_as<bool>; } is true.

Returns: A mask_type object initialized with the results of applying *op* to lhs and rhs as a binary element-wise operation.

9.9. Modify [simd.unary]

Modify the constraints in [simd.unary] as follows:

Let *op* be the operator.

Constraints: ~~requires (value_type a) { op a; } is true~~ supported-unary-op<value_type, Op> is true, where Op is the corresponding standard transparent function object (negate<>, bit_not<>).

Returns: A basic_simd object initialized with the results of applying *op* to v as a unary element-wise operation.

9.10. Feature test macro [version.syn]

Add to [version.syn]:

```
#define __cpp_lib_simd_udt YYYYMMML // also in <simd>
```

9.11. (OPTIONAL) Add overload for to_underlying

Note: This wording is included if the committee chooses to adopt the utility functions from the Design Options section.

Add to [simd.cast]:

```
template<class Enum, class Abi>
constexpr rebind_t<underlying_type_t<Enum>, basic_simd<Enum, Abi>>
    to_underlying(const basic_simd<Enum, Abi>& v) noexcept;
```

Constraints: `is_enum_v<Enum>` is true.

Returns: A `basic_simd` object where element `i` is `to_underlying(v[i])`.

§ 9.12. (OPTIONAL) Add overload for `to_integer`

Note: This wording is included if the committee chooses to adopt the utility functions from the Design Options section.

Add to [simd.cast]:

```
template<class IntegerType, class Abi>
constexpr rebind_t<IntegerType, basic_simd<byte, Abi>>
    to_integer(const basic_simd<byte, Abi>& v) noexcept;
```

Constraints: `is_integral_v<IntegerType>` is true.

Returns: A `basic_simd` object where element `i` is `to_integer<IntegerType>(v[i])`.

§ 10. Conclusion

This proposal extends `std::simd` to support user-defined element types through a minimal, principled change where the closed list of vectorizable types is replaced with trait-based constraints.

Earlier revisions explored explicit customization mechanisms, leading to complicated designs. Committee feedback encouraged exploring element-wise inference. The working draft specification already defines all operations through element-wise application, so changing only the definition of which types are allowed provides the extension we need.

Committee discussion raised legitimate concerns about whether compilers could actually optimize user-defined operator calls into efficient vector code. Implementation experience with leading compilers (Clang 20, Intel oneAPI 2025.0) has shown that they can. While compiler maturity varies across vendors and versions, the results demonstrate the fundamental viability of the element-wise inference approach.

By changing only the gate-keeping logic for vectorizable types, we enable type safety for strong typedefs, domain-specific types for signal processing and other specialized domains, enumerations, `std::byte`, and small compound types. This is achieved with no breaking changes to existing code and no modification to any operation semantics.

Implementation experience identifies opportunities for future customization mechanisms where performance tuning might be valuable, which we present as a design alternative for committee consideration. However, customization is not essential for functionality or reasonable performance.

§ 11. Acknowledgements

We would like to thank Matthias Kretz for his feedback and contributions to discussions throughout the development of this proposal. We also thank the members of SG1 and SG6 who provided feedback during recent meetings, which significantly shaped the direction of this revision.

§ 12. Appendix: Customization Point Technical Details

This appendix provides technical details for the ADL-based customization mechanism proposed in [§ 7 Design Alternative: Customization Points](#).

§ 12.1. Dual Dispatch Strategy

The customization design uses separate code paths based on type category:

```
// Arithmetic types, std::byte, std::complex: always optimized
template<typename T>
    requires std::is_arithmetic_v<T> || std::is_same_v<T, std::byte> || /* complex */
friend constexpr basic_simd operator+(const basic_simd& lhs, const basic_simd& rhs)
{
    return /* implementation-defined optimized implementation */;
}

// Enumerations and user-defined types: check for customization via ADL
template<typename T>
    requires (!std::is_arithmetic_v<T> && !std::is_same_v<T, std::byte> && /* not complex */)
friend constexpr basic_simd operator+(const basic_simd& lhs, const basic_simd& rhs)
    requires requires (value_type a, value_type b) { { a + b } -> std::same_as<value_type>; }
{
    if constexpr (requires { simd_operator(lhs, rhs, std::plus<>{}); }) {
        return simd_operator(lhs, rhs, std::plus<>{}); // Custom via ADL
    } else {
        return /* element-wise application */; // Default
    }
}
```

This ensures:

- **Arithmetic types, `std::byte`, `std::complex`:** Always optimized, never check for customization
- **Enumerations:** Can provide `simd_operator` customization if they have custom operators; otherwise element-wise inference produces optimal code for standard enum operations
- **User-defined types:** Optional customization with element-wise fallback
- **Performance guarantee:** No overhead for standard arithmetic types

Users control their own target-specific optimizations if desired:

```
// User code for target-specific optimization
namespace my_lib {
    enum class PackedColor : uint32_t { /* ... */ };

    // Custom enum operator
    PackedColor operator+(PackedColor a, PackedColor b) {
        return /* custom blending logic */;
    }

    // Optional SIMD optimization
    auto simd_operator(vec<PackedColor> lhs, vec<PackedColor> rhs, std::plus<>) {
        #ifdef __AVX512F__
            return my_avx512_blend(lhs, rhs);
        #else
            return my_generic_blend(lhs, rhs);
        #endif
    }
}
```


12.2. Complete Example with Selective Customization

This example shows how users can customize specific operations while relying on element-wise inference for others:

```
namespace my_lib {
    struct fixed_point_16s8 {
        std::int16_t data;

        // Basic operators use normal semantics
        fixed_point_16s8 operator+(fixed_point_16s8 rhs) const {
            return fixed_point_16s8{data + rhs.data};
        }

        fixed_point_16s8 operator-(fixed_point_16s8 rhs) const {
            return fixed_point_16s8{data - rhs.data};
        }

        bool operator<(fixed_point_16s8 rhs) const {
            return data < rhs.data;
        }
    };

    // Customize multiply (requires scaling) - Binary operation
    template<typename Abi>
    auto simd_operator(
        const basic_vec<fixed_point_16s8, Abi>& lhs,
        const basic_vec<fixed_point_16s8, Abi>& rhs,
        std::multiplies<>)
    {
        // Custom implementation with appropriate scaling
        // Could use intrinsics or library functions
        return /* optimized multiply with scaling */;
    }

    // Customize divide (requires scaling) - Binary operation
    template<typename Abi>
    auto simd_operator(
        const basic_vec<fixed_point_16s8, Abi>& lhs,
        const basic_vec<fixed_point_16s8, Abi>& rhs,
        std::divides<>)
    {
        // Custom implementation with appropriate scaling
        return /* optimized divide with scaling */;
    }

    // Addition, subtraction, comparisons use element-wise inference
    // No customization needed for these simple operations
}

// Usage
vec<my_lib::fixed_point_16s8> a, b;
auto sum = a + b;      // Uses element-wise inference (fast)
auto diff = a - b;     // Uses element-wise inference (fast)
auto product = a * b;  // Uses custom simd_operator (optimal)
auto quotient = a / b; // Uses custom simd_operator (optimal)
auto mask = a < b;     // Uses element-wise inference (fast)
```

Conversion example:

```
namespace my_lib {
    struct BFloat16 { uint16_t bits; /* ... */ };

    // Optimize conversion to float
    template<typename Abi>
```

```

basic_vec<float, Abi>
simd_convert(const basic_vec<BFloat16, Abi>& source, convert_to_t<float>) {
    // Use hardware bfloat16 conversion if available
    #ifdef __AVX512BF16__
        return /* use vcvtneps2bf16 or similar */;
    #else
        return /* shift bits implementation */;
    #endif
}
}

```

This demonstrates the key benefit: users customize only what needs optimization while relying on inference for everything else. The single `simd_operator` name handles unary, binary, and ternary operations through overloading.

§ 13. Appendix: Assembly Code Examples

This section provides detailed assembly listings from the implementation experience, demonstrating how element-wise inference generates optimal vector code. Testing was performed with Clang 20 and Intel oneAPI 2025.0 targeting Intel Sapphire Rapids.

Complex Expression Composition

Element-wise operations compose well across multiple operations in a single expression:

C++ Code	Generated Assembly
<pre> // Strong typedef auto compute(vec<Meters> a, vec<Meters> b, vec<Meters> c) { return (a + b) * c - a; } // Built-in type (for comparison) auto compute(vec<float> a, vec<float> b, vec<float> c) { return (a + b) * c - a; } </pre>	<pre> ; Strong typedef Meters compute(...): vaddps zmm1, zmm1, zmm0 vfmsub231ps zmm0, zmm2, zmm1 ret ; Built-in float compute(...): vaddps zmm1, zmm1, zmm0 vfmsub231ps zmm0, zmm2, zmm1 ret </pre>

The assembly is identical for both the user-defined type and the built-in type, demonstrating that user-defined types achieve zero-overhead abstraction. The compiler successfully fuses multiple operations and optimizes register allocation regardless of whether the element type is `Meters` or `float`.

C++ Code	Generated Assembly
<pre> auto broadcast(int16_t x) { return vec<saturating_int16>(x); } </pre>	<pre> broadcast(short): vpbroadcastw zmm0, edi ret </pre>
<pre> auto iq_swap(const vec<saturating_int16>& v) { return permute(v, [](auto idx) { return idx ^ 1; }); } </pre>	<pre> iq_swap(...): vprold zmm0, zmmword ptr [rdi], 1 ret </pre>
<pre> auto add(vec<saturating_int16> lhs, vec<saturating_int16> rhs) { </pre>	<pre> add(...): vpaddsw ymm0, ymm0, ymm1 ret </pre>

C++ Code	Generated Assembly
<pre>return lhs + rhs; }</pre>	
<pre>auto compound_add(vec<saturating_int16> lhs, vec<saturating_int16> rhs) { lhs += rhs; return lhs; }</pre>	<pre>compound_add(...): vpaddsw ymm0, ymm0, ymm1 ret</pre>
<pre>auto cmp_gt(vec<saturating_int16> lhs, vec<saturating_int16> rhs) { return lhs > rhs; }</pre>	<pre>cmp_gt(...): vpcmpgtw ymm0, ymm1, ymm0 ret</pre>
<pre>auto biggest(vec<saturating_int16> lhs, vec<saturating_int16> rhs) { return max(lhs, rhs); }</pre>	<pre>biggest(...): vpmasw ymm0, ymm0, ymm1 ret</pre>
<pre>auto distance(vec<Meters, 8> x, vec<Meters, 8> y) { return x + y; }</pre>	<pre>distance(...) vaddps ymm0, ymm0, ymm1 ret</pre>
<pre>auto closer(vec<Meters> x, vec<Meters> y) { return x < y; }</pre>	<pre>closer(...) vcmpltb ymm0, ymm0, ymm1 ret</pre>
<pre>auto dimmer(vec<Color> x, vec<Color> y) { return x < y; }</pre>	<pre>dimmer(...) vpcmpgtd ymm0, ymm1, ymm1 ret</pre>
<pre>auto load_and_convert(std::span<short, 1024> s) { return unchecked_load<vec<Meters, 8>>(s); }</pre>	<pre>load_and_convert(...): # vcvtqdq2ps ymm0, ymmword ret</pre>
<pre>auto gather(std::span<int, 1024> s, const vec<int, 8> indexes) { return unchecked_gather_from<vec<Meters, 8>>(s, indexes); }</pre>	<pre>gather(): # kxnorw k1, k0, k0 vpxor xmm1, xmm1, xmm1 vpgatherdd ymm1 {k1}, ymm0 vcvtqdq2ps ymm0, ymm1 ret</pre>

These examples demonstrate optimal code generation with native vector instructions and no scalar fallback.

§ References

§ Informative References

[P4006]

Daniel Towner. *Transparent wrappers for shift operators*. URL: <https://wg21.link/P4006>

[SIMD.GENERAL]

General requirements for SIMD types. URL: <https://eel.is/c++draft/simd#general>